James B. Ponwith

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Principal Software Engineer and technical lead specializing in distributed systems, resiliency, and cloud infrastructure. Architect of a multi-region platform used by 15+ teams across 3 orgs, safeguarding ~$780B in processing at a 99.99% SLA.

PROFESSIONAL EXPERIENCE

Capital One (NYSE: COF) **May 2022 – Present**

Principal Software Engineer | Technical Lead – Card Tech Authorizations (Promoted from SE and Senior SE)

- Primary architect and technical owner of a multi-region resiliency platform, growing a greenfield Go service into a 28-repository ecosystem serving 8 tenant applications and 15+ teams across 3 orgs
- Tech lead (1.5 yrs), acting eng. manager (6 mos): set direction for 15 engineers; mentor 8; brief EVP/SVP
- Designed a highly concurrent Go Resiliency Engine that auto-detects failures and routes live traffic to healthy cells, cutting MTTR 95% to ~1 minute and protecting $2M+ net income/yr across systems processing ~$780B
- Engineered strict cross-region consistency: DynamoDB MRSC, conditional writes, peer-to-peer lease protocol
- Led multi-ingress traffic-management architecture (CloudWatch→Route 53 failover, direct MQ control, custom routing) and the migration to AWS ARC, shielding 20+ services with zero Sev-3+ incidents over the past year
- Own on-call and incident response for the platform: operational OKRs of MTTD/MTTC <5 min and Sev4+ recovery <2 h; OpenTelemetry alerting/dashboards and fault-injection drills eliminate recurring failure modes
- Built low-latency Kafka-based integrations for cross-domain services on ECS (dynamic S3 config, OpenTelemetry)
- Built the dev/test paved path: reproducible local and ephemeral environments (LocalStack Pro AWS emulation) for unit, component, live-dependency, and fault-injection suites, so engineers and agents validate changes before merge
- Built and scaled an agent-driven delivery workflow (intent → build → validate → release → learn) across 28 repos: 100% of PRs agent-written and human-reviewed, PR cycle time cut 90% (10 days → 1), test coverage 5% → 95%
- Standardized agentic workflows org-wide (30 engineers on 3 outside teams; 120+ trained) and extended them with async agents that monitor PRs and CI/CD, review, patch failures, and re-run validation behind human merge review
- Operate MCP tooling for agents (GitHub, Jira, Slack, enterprise): tool/model allowlists, audit logs, spend controls

Qualcomm (NASDAQ: QCOM) **August 2019 – May 2022**

SVE Engineer – Qualcomm CDMA Technologies | Systems Validation and Emulation

- Built automation and visualization pipelines that standardized systems validation, reducing triage time by 90%

All About Tennis **2020 – Present**

Lead Developer and Computer Systems Manager – Family Retail Business (part-time)

- Modernized retail systems during COVID, unifying in-store and online sales; ~700% YoY ecommerce growth for 2 years

OPEN SOURCE

phux (co-founder; Rust, Swift, Kotlin): terminal multiplexer shared by humans and AI agents; built foundational protocol features (relay routing, cross-session pane moves, remote auth & encryption), and the iOS/Android app
llm-resiliency-router (Go): LLM gateway with health cells, mid-stream failover, hedging, canary evals, and chaos profiles
agentic-flywheel (author): Go fleet coordinator running agents async overnight across 5 repos: lease-based work claims with heartbeats, review-budget allocation, worktree isolation, fleet-wide kill switch • Contributor: blackbird, TerminalPhone

TECHNICAL SKILLS

Languages: Go, Rust, Python, C/C++, Swift, Kotlin, TypeScript/JavaScript, SQL, Bash
AWS: ECS, Lambda, DynamoDB, S3, SQS, SNS, Route 53, ARC, CloudFormation, IAM, KMS, Secrets Manager, API Gateway
Data & Messaging: DynamoDB, MySQL, Kafka, IBM MQ, gRPC, REST, mTLS
Distributed Systems: active-active multi-region, cell routing, circuit breaking, load shedding, failover/DR, distributed leases
Infra & Delivery: Docker, Kubernetes, Helm, Terraform, Vault, GitHub Actions, ArgoCD, Jenkins, Godog, LocalStack Pro
Observability: OpenTelemetry, New Relic, Observe, Datadog, Splunk, CloudWatch, PagerDuty
Agentic AI: Claude Code, OpenCode, pi, beads, OpenSpec, Blackbird (MCP), multi-agent orchestration, LLM APIs & evals

EDUCATION & ADDITIONAL

University of San Diego: BA Computer Science, *School of Engineering* • BBA Finance, *School of Business* **May 2019**
Earlier roles: SWE Intern, IAC • Network Engineer, Pacifica Real Estate • VC Associate, University Growth Fund (2017–2019)
NCAA Division I Tennis — 2× WCC Champion • Capital One Tech Excellence Award